

An Evolutionary Algorithm for Actuator-Sensor-Communication Co-Design in Distributed Control

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Abstract—This paper studies the co-design of actuators, sensors, and communication in the distributed setting, where a networked plant is partitioned into subsystems each equipped with a sub-controller interacting with other sub-controllers. The objective is to jointly minimize control cost (measured by LQ cost) and material cost (measured by the number of actuators, sensors, and communication links used). We approach this using an evolutionary algorithm to selectively prune a baseline dense LQR controller. We provide convergence and stability analyses for this algorithm. For unstable plants, controller pruning is more likely to induce instability; we provide an algorithm modification to address this. The proposed method is validated in simulations. One key result is that co-design of a 98-state swing equation model can be done on a standard laptop in seconds; the co-design outperforms naive controller pruning by over 50%.

I. INTRODUCTION & MOTIVATION

Large-scale networked dynamical systems arise in a wide range of modern engineering applications, including power grids, traffic networks, multi-agent robotic systems, and cyber-physical infrastructures. In such systems, centralized control architectures are often impractical due to scalability limitations and communication constraints. These challenges have motivated extensive research on *distributed control*, where control actions are computed using only local information and limited inter-agent communication [1]–[3]. There is also increasing interest in *co-design* methodologies that jointly consider control performance, sensor and actuator placement, and communication structure. In the distributed control setting, co-design remains a fundamentally difficult problem, often translating to nonlinear mixed-integer programming problems (NLMIPs) or other similarly combinatorial problems. Existing approaches often focus on special subsets of the co-design problem which can be reduced to a more tractable form [4]–[7]. However, this reduction eliminates large portions of the design space, which may contain more cost-effective solutions.

In this paper, we study the co-design of actuators, sensors, and communication in the distributed linear-quadratic (LQ) setting via evolutionary algorithms (EA). EAs are heuristic optimization methods inspired by natural selection, and have demonstrated efficacy on complex problems such as NLMIPs [8], [9]. The problem setup is provided in Section II; the EA approach is described in Section III. We then provide convergence (Section IV) and stability (Section V) analysis, as well as an additional sparsity-promoting modification to

the base EA (Section VI); our approach and analyses are validated in simulation (Section VII).

II. PROBLEM SETUP

Consider a discrete-time linear time-invariant system:

$$x_{k+1} = Ax_k + Bu_k \quad (1)$$

where $x_k \in \mathbb{R}^{N_x}$ is the state vector, $u_k \in \mathbb{R}^{N_u}$ is the control input, $A \in \mathbb{R}^{N_x \times N_x}$ is the state matrix, and $B \in \mathbb{R}^{N_x \times N_u}$ is the input matrix. The system contains N interconnected subsystems, each having one or more states. State and control are partitioned into $[x]_i$, $[u]_i$, and $[w]_i$ for each subsystem i ; system matrices A and B are partitioned into $[A]_{ij}$, $[B]_{ij}$, which capture subsystem-level interaction. System topology is described by an unweighted directed graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$, where vertex v_i corresponds to subsystem i , and edge $e_{ij} \in \mathcal{E}$ exists whenever $[A]_{ij} \neq 0$ or $[B]_{ij} \neq 0$. Let $\mathcal{A} \in \{0, 1\}^{N \times N}$ be the adjacency matrix for this graph, where by convention $\mathcal{A}_{ii} = 0$. For a state-feedback controller $u_k = Kx_k$, we can also partition K into subsystems. Here, row $[K]_{i,:}$ is the sub-controller at subsystem i , and $[K]_{ij}$ represents information required by sub-controller i from sub-controller j . We can similarly build a controller adjacency matrix $\mathcal{K}(K)$, where $\mathcal{K}(K)_{ij} = 1$ wherever $[K]_{ij} \neq 0$, and $\mathcal{K}_{ii} = 0$ by convention.

The number of actuators used by the controller is $N_a(K) := \|K\|_{1,0}$, i.e., the number of nonzero rows in K . Similarly, the number of sensors used by the controller is $N_s(K) := \|K\|_{0,1}$, i.e., the number of nonzero columns in K . The number of (inter-subsystem) communication links used by the controller can be written as $N_c(K) = \|\mathcal{K}(K)\|_0$, i.e., the number of nonzeros in the controller adjacency matrix.

The high-level co-design problem is:

$$\begin{aligned} \min_K \quad & J(K) + w_a N_a(K) + w_s N_s(K) + w_c N_c(K) \\ \text{s.t.} \quad & (1), u_k = Kx_k \end{aligned} \quad (2)$$

where J is some control objective, and w_a , w_s , and w_c are scalar penalties on actuator, sensor, and communication link usage. In other words, we seek a state feedback controller K for plant (1) that jointly minimizes the control objective as well as material costs required by this controller.

In this paper, we will consider the case of co-designing a linear quadratic regulator (LQR) via pruning; this is inspired by previous work on the structure and prune-ability of dense LQR controllers [3]. Specifically, given state and penalty matrices Q and R , we first synthesize an optimal

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LQR controller K_d (which is generally dense), then set select entries in K_d to zero. We now reparameterize (2) to reflect this. We first define the optimization vector

$$\theta = [\ell, \mathbf{a}, \mathbf{s}] \quad (3)$$

where link count $\ell \in [1, N_u N_x]$ is an integer determining the number of nonzero elements to keep from K_d , actuator mask $\mathbf{a} \in \{0, 1\}^{N_u}$ is a binary vector representing actuator selection, and sensor mask $\mathbf{s} \in \{0, 1\}^{N_x}$ is a binary vector representing sensor selection.

From parameter θ and dense controller K_d , we obtain the pruned (i.e., sparsified) controller $K_s(\theta)$. To do so, first sort the nonzero elements of K_d by magnitude; keep the first ℓ elements, and set the rest to zero. Let $\Pi_\ell(K_d)$ denote this process. Then, let $\bar{\mathbf{a}}$ denote $\text{not}(\mathbf{a})$, i.e., $\mathbf{a}$ with ones and zeros flipped (and similarly for $\bar{\mathbf{s}}$). Set rows $K_{\bar{\mathbf{a}}\cdot} = 0$, then set columns $K_{\cdot\bar{\mathbf{s}}} = 0$. Let $\Pi_{\mathbf{a},\mathbf{s}}(K_d)$ denote this process.

We choose control objective $J(K) = \frac{J_{\text{LQR}}(K)}{J_{\text{LQR}}(K_d)}$, i.e., the LQ performance of controller K relative to the optimal dense controller. Cost $J_{\text{LQR}}(K)$ is

$$J_{\text{LQR}}(K) = \mathbb{E}_{x_0 \sim \mathcal{N}(0, \Sigma)} \left[\sum_{k=0}^{\infty} (x_k^\top Q x_k + u_k^\top R u_k) \right] \quad (4)$$

under dynamics (1), $u_k = K x_k$

This objective is infinite when closed-loop $A + BK$ is unstable. When the closed-loop is stable, the objective can be evaluated by solving the discrete-time Lyapunov equation for the closed-loop system $A + BK$ to obtain a matrix P , then taking the trace of $P\Sigma$. This cost goes to infinity when the closed-loop system is unstable. The new co-design problem can be written as:

$$\begin{aligned} \min_{\theta} \quad & J_{\text{EA}}(\theta) \\ \text{where} \quad & J_{\text{EA}}(\theta) = \frac{J_{\text{LQR}}(K_s(\theta))}{J_{\text{LQR}}(K_d)} + w_a N_a(K_s(\theta)) \\ & + w_s N_s(K_s(\theta)) + w_c N_c(K_s(\theta)) \end{aligned} \quad (5)$$

Notation. $\|\cdot\|$ denotes the induced 2-norm for matrices and the Euclidean norm for vectors; $\|\cdot\|_F$ denotes the Frobenius norm.

III. EVOLUTIONARY ALGORITHM FOR CO-DESIGN

We now propose an evolutionary algorithm (EA) to solve (5). EAs work upon a population of individuals with population size N_p . Each individual i is characterized by its gene (i.e., parameter) θ^i . The population at generation t is written as $\mathcal{P}_t := \{\theta^1, \theta^2, \dots, \theta^{N_p}\}$. Our population $\mathcal{P}_0$ is initialized with randomly generated individuals; for each individual, we set $\ell = \|K_d\|_0$, and randomly draw elements in $\mathbf{a}$ and $\mathbf{s}$ from a Bernoulli distribution. Then, at each generation t , we evaluate the cost $J_{\text{EA}}(\theta^i)$ of each individual θ^i in population $\mathcal{P}_t$. The n_e individuals with the lowest cost are directly carried over to the next generation's population $\mathcal{P}_{t+1}$. The remainder of the next generation's population are generated using the following operations:

Algorithm 1 EA-Based Sparse LQR Controller Co-Design

Input:

Plant matrices A, B
Objective parameters Q, R, w_c, w_a, w_s
EA parameters $N_p, G_{\max}, p_c, p_m, n_e, \tau, d$

- 1: Compute optimal LQR controller K_d
 - 2: Randomly generate $\mathcal{P}_0 = \{\theta^i\}_{i=1}^{N_p}$
 - 3: **for** $t = 1$ **to** $G_{\max}$:
 - 4: **for** each $\theta^i \in \mathcal{P}_{t-1}$:
 - 5: Evaluate $J_{\text{EA}}(\theta^i)$
 - 6: $\mathcal{P}_t \leftarrow \{n_e \text{ lowest-cost individuals from } \mathcal{P}_{t-1}\}$
 - 7: **for** $j = n_e + 1$ **to** N_p :
 - 8: $\theta^{p1}, \theta^{p2} \leftarrow \text{Selection}(\mathcal{P}_{t-1}, \tau)$
 - 9: $\theta^c \leftarrow \text{Crossover}(\theta^{p1}, \theta^{p2}, p_c)$
 - 10: $\theta^c \leftarrow \text{Mutation}(\theta, p_m, d)$
 - 11: $\mathcal{P}_t \leftarrow \mathcal{P}_t \cup \theta^c$
 - 12: $\theta^* \leftarrow \text{argmin}_{\theta^i \in \mathcal{P}_t} J_{\text{EA}}(\theta^i)$
- Return:** $K_s(\theta^*)$

- 1) **Selection:** The operator $\text{Selection}(\mathcal{P}_t, \tau)$ samples two distinct individuals (i.e., parents) from population $\mathcal{P}_t$. The probability of individual i being chosen is

$$P(i) = \frac{\exp(-J_{\text{EA}}(\theta^i)/\tau)}{\sum_{\theta^j \in \mathcal{P}_t} \exp(-J_{\text{EA}}(\theta^j)/\tau)} \quad (6)$$

where τ is the temperature parameter. Individuals with lower cost are more likely to be chosen.

- 2) **Crossover:** The operator $\text{Crossover}(\theta^{p1}, \theta^{p2}, p_c)$ takes genes from two distinct individuals (i.e., parents) and crossover probability p_c , and generates a child gene. First, draw k from uniform distribution $\mathcal{U}(1, N_\theta - 1)$ where N_θ denotes the length of the gene vector. Then, generate the child gene θ^c as $\theta^c = [\theta_{1:k}^{p1}, \theta_{k+1:N_\theta}^{p2}]$.
- 3) **Mutation:** The operator $\text{Mutation}(\theta, p_m, d)$ takes gene $\theta = [\ell, \mathbf{a}, \mathbf{s}]$, mutation probability p_m , and mutation range d , and generates a mutated gene θ^m . Define mutation vectors $\mathbf{m}_a$ and $\mathbf{m}_s$, where each element m_j of these vectors are drawn from Bernoulli(p_m). Then, define mutation scalar δ which is drawn from $\text{Unif}\{-d, d\}$. The mutated gene is $\theta^m = [\text{sat}(\ell_c + \delta), \mathbf{a} \oplus \mathbf{m}_a, \mathbf{s} \oplus \mathbf{m}_s]$, where $\text{sat}(\cdot)$ clips the value to interval $[1, N_u N_x]$, and $\oplus$ denotes the element-wise XOR operation.

$N_p - n_e$ pairs of parents are selected to reproduce via crossover and mutation; the resulting children are added to the subsequent population. This process repeats until the maximum number of generations $G_{\max}$ is reached. At this point, the gene of best-performing individual, denoted θ^* , is used as the EA solution to (5). The overall EA is summarized in Algorithm 1. The complexity of the algorithm is $\mathcal{O}(G_{\max} N_p N_x^3)$; this is dominated by cost analysis, which requires either eigenvalue computation or solving a Lyapunov equation for each individual in each generation, both of which have cubic complexity.

We demonstrate the efficacy of EA in simulations in

Section VII. Now, we discuss the convergence and stability properties of this algorithm.

IV. CONVERGENCE OF EA CO-DESIGN

We provide an approximate analysis for the convergence of the EA. This will be done by leveraging properties of LQR truncations previously derived in [3]. First, we introduce some relevant definitions and assumptions.

Definition 1. Let $L > 0$ and $\alpha \in [0, 1)$. Then,

- (a) Φ is (L, α) -stable if $\|\Phi^k\| \leq L\alpha^k \quad \forall k \geq 0$.
- (b) (A, B) is (L, α) -stabilizable if $\exists K$ such that $\|K\| \leq L$ and $A + BK$ is (L, α) -stable.
- (c) (A, C) is (L, α) -detectable if $(A^\top, C^\top)$ is (L, α) -stabilizable.

Assumption 1. $\exists L > 1$, $\alpha \in (0, 1)$, and $\gamma \in (0, 1)$ such that:

- (a) $\|A\|, \|B\|, \|Q\|, \|R\| \leq L$
- (b) $R \succeq \gamma I$
- (c) (A, B) is (L, α) -stabilizable
- (d) $Q \succeq 0$, and $(A, Q^{1/2})$ is (L, α) -detectable.

These are inherited from [3] and will be assumed to hold for the remainder of the paper. We note that Assumption 1 is merely a more precise statement of the standard assumption of stabilizability for linear systems. Now, we recall the following result from [3] on the spatial decay of the optimal LQR gain:

Lemma 1. Let $d_G(i, j)$ denote the number of edges in the shortest path between nodes i and j in the system graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Then, optimal LQR gain K_d satisfies $\|K_{d,ij}\| \leq \Upsilon \rho^{d_G(i,j)}$, where ρ is a constant in $(0, 1)$ and Υ is a constant lower-bounded by 1.

Values for ρ and Υ are provided in [3]; for our purposes, it suffices to know their existence. Now, we recall a result from [10] on the Lipschitz continuity of LQR cost:

Lemma 2. For every sublevel set $\mathcal{S} = \{K : J_{\text{LQR}}(K) \leq c, \}$, $\exists L_J > 0$ such that for all $K_1, K_2 \in \mathcal{S}$,

$$\|\nabla_K J_{\text{LQR}}(K_1) - \nabla_K J_{\text{LQR}}(K_2)\|_F \leq L_J \|K_1 - K_2\|_F \quad (7)$$

By definition, $K_d \in \mathcal{S}$ is a first-order stationary point of the LQR objective, i.e., $\nabla_K J_{\text{LQR}}(K_d) = 0$. Then $\forall K \in \mathcal{S}$,

$$J_{\text{LQR}}(K) - J_{\text{LQR}}(K_d) \leq \frac{L_J}{2} \|K - K_d\|_F^2. \quad (8)$$

We now proceed with convergence analysis. For analytical simplicity, we choose to focus only on two features of the optimization objective (5): the control objective $J(K) = \frac{J_{\text{LQR}}(K)}{J_{\text{LQR}}(K_d)}$ and the communication co-design objective $w_c N_c(K_s(\theta))$. For the remainder of this section, we assume $\mathbf{a} = \mathbf{1}$ and $\mathbf{s} = \mathbf{1}$ where $\mathbf{1}$ is the ones vector, i.e., all actuators and sensors are retained. Pruning (i.e., sparsifying) K typically has opposite effects on the two objectives we consider; it increases $J(K)$ while decreasing co-design costs. In simulation (Section VII), we see that the provided bound is quite close to the true convergence rate even when the EA

uses the full optimization objective with $\mathbf{a}$ and $\mathbf{s}$. We now define some additional EA-related quantities.

Definition 2. For each generation t , define

- (a) Best individual θ_t^* , where $\theta_t^* := \operatorname{argmin}_{\theta \in \mathcal{P}_t} F(\theta)$, with link count ℓ_t^* and controller $K_t := K_s(\theta_t^*)$.
- (b) Link reduction X_t , where for $t \geq 1$,

$$X_t := \ell_{t-1}^* - \ell_t^*, \quad |X_t| \leq d. \quad (9)$$

- (c) Effective truncation distance h_t . Define simplified gene $\hat{\theta} := [\ell_t^*, \mathbf{1}^\top, \mathbf{1}^\top]$, and build controller adjacency matrix $\mathcal{K}(K_s(\hat{\theta}))$ with edges $\mathcal{E}_\mathcal{K}$. Then, h_t is the largest integer such that for all edges (i, j) with distance $d_G(i, j) \leq h_t$, $(i, j) \in \mathcal{E}_\mathcal{K}$.

Note that by construction, lowest-cost individual in each generation of an EA will always carry over into the next generation, so $\theta_t^* \in \mathcal{P}_{t+1}$ and $J_{\text{EA}}(\theta_t^*) \geq J_{\text{EA}}(\theta_{t+1}^*)$. We now present a series of results that are required to get to the final convergence result (Theorem 1).

Lemma 3. Define the cost-rate function

$$\Phi(h) := \frac{L_J \Upsilon^2 \rho^{2h}}{J_{\text{LQR}}(K_d)} \left(\sqrt{N_u N_x} + \frac{1}{2} \right) \quad (10)$$

and the critical truncation distance

$$h^* := \left\lceil \frac{1}{2|\log \rho|} \log \frac{L_J \Upsilon^2 (\sqrt{N_u N_x} + \frac{1}{2})}{w_c J_{\text{LQR}}(K_d)} \right\rceil. \quad (11)$$

Then, when $h_t > h^*$, $\Phi(h_t) < w_c$ and pruning additional link(s) yields a net decrease in J_{EA} . Conversely, when $h_t \leq h^*$, pruning additional link(s) does not decrease J_{EA} . The proof is in 1.

Lemma 4. Define $K_{t-1} := K_s(\theta_{t-1}^*)$ and $K_t := K_s(\theta_t^*)$. If $X_t \geq 1$, then,

$$\frac{J_{\text{LQR}}(K_t) - J_{\text{LQR}}(K_{t-1})}{J_{\text{LQR}}(K_d)} \leq \Phi(h_{t-1}) X_t. \quad (12)$$

Proof. By Lemma 2, we can choose constant L_J such that

$$\begin{aligned} J_{\text{LQR}}(K_t) - J_{\text{LQR}}(K_{t-1}) &\leq \langle \nabla_K J(K_{t-1}), K_t - K_{t-1} \rangle + \frac{L_J}{2} \|K_t - K_{t-1}\|_F^2. \end{aligned} \quad (13)$$

Since $\operatorname{supp}(K_t) \subset \operatorname{supp}(K_{t-1})$, the X_t removed entries each satisfy $|K_d(i, j)| \leq \Upsilon \rho^{h_{t-1}}$ by Lemma 1, so

$$\|K_t - K_{t-1}\|_F \leq \sqrt{X_t} \Upsilon \rho^{h_{t-1}}. \quad (14)$$

For the gradient term, $\nabla J(K_d) = 0$ gives

$$\|\nabla J(K_{t-1})\|_F \leq L_J \|K_{t-1} - K_d\|_F \leq L_J \sqrt{N_u N_x} \Upsilon \rho^{h_{t-1}}.$$

Combining via Cauchy-Schwarz and using $\sqrt{X_t} \leq X_t$ for $X_t \geq 1$:

$$\begin{aligned} J_{\text{LQR}}(K_t) - J_{\text{LQR}}(K_{t-1}) &\leq L_J \Upsilon^2 \rho^{2h_{t-1}} \left(\sqrt{N_u N_x} + \frac{1}{2} \right) X_t. \end{aligned}$$

Dividing by $J_{\text{LQR}}(K_d)$ obtains $\Phi(h_{t-1}) X_t$. $\square$

Proposition 1. If $X_t \geq 1$ and $h_{t-1} > h^*$, then

$$J_{\text{EA}}(\theta_{t-1}^*) - J_{\text{EA}}(\theta_t^*) \geq (w_c - \Phi(h_{t-1})) X_t > 0. \quad (15)$$

In particular, h^* is obtained by solving $\Phi(h) = w_c$ for h , which yields (11).

Proof. From Lemma 2, we have

$$J_{\text{EA}}(\theta_{t-1}^*) - J_{\text{EA}}(\theta_t^*) = w_c X_t - \frac{J_{\text{LQR}}(K_t) - J_{\text{LQR}}(K_{t-1})}{J_{\text{LQR}}(K_d)}.$$

By Lemma 4, the second term is at most $\Phi(h_{t-1}) X_t$. By Definition 2, $h_{t-1} > h^*$ implies $\Phi(h_{t-1}) < w_c$. $\square$

Proposition 2 (Improvement probability). For each generation t with $h_{t-1} > h^*$, the probability that a single offspring strictly improves J_{EA} satisfies

$$p_{\text{imp}}(t) \geq \frac{(1 - p_m)^{N_u + N_x}}{N_p (2d + 1)} \mathbb{P}[K_t \in \mathcal{S}]. \quad (16)$$

Proof. We construct an explicit improvement path: selection chooses θ_{t-1}^* as first parent ($\geq 1/N_p$); crossover preserves ℓ_{t-1}^* (ℓ at position 1, split $k \geq 1$); mutation sets $\delta = -1/(2d+1)$ with $\mathbf{a}, \mathbf{s}$ unchanged ($(1 - p_m)^{N_u + N_x}$). The offspring has $X_t = 1$ and identical masks. Proposition 1 applies provided $K_t \in \mathcal{S}$; a sufficient condition is $\sqrt{N_u N_x} \Upsilon \rho^{h(\ell_{t-1}^*)} < \sigma_{\text{crit}}$, which is quantified in Section V (Theorem 2). $\square$

Finally, we present the theorem on convergence:

Theorem 1. Let $\Phi(\cdot)$ and h^* be as defined in Lemma 3. For generation $t \geq 1$, define per-offspring improvement probability

$$p_{\text{imp}}(t) := \frac{1}{N_p (2d + 1)} \mathbf{1}\{h_{t-1} > h^*\}, \quad (17)$$

where $\mathbf{1}$ is the indicator function. Define the population-level improvement probability as $P_{\text{imp}}(t) := 1 - (1 - p_{\text{imp}}(t))^{N_p - n_e}$. Then, the following per-generation improvement lower bound holds: $\forall t$ with $h_{t-1} > h^*$,

$$\mathbb{E}[J_{\text{EA}}(\theta_{t-1}^*) - J_{\text{EA}}(\theta_t^*)] \geq (w_c - \Phi(h_{t-1})) P_{\text{imp}}(t) > 0. \quad (18)$$

(ii) Guaranteed pruning depth. For all t with $h_t > h^*$, the EA continues to prune with positive probability. In particular, the terminal link count satisfies $\ell_\infty^* \leq h^{-1}(h^*)$.

Proof. First, combine Lemmas 2 and 1 to bound the per-step LQR cost increase. When $h_{t-1} > h^*$, the EA cost increase due to LQR cost increase LQR is outweighed by the EA cost decrease due to the reduction in communication links ($w_c X_t$), yielding a net decrease in EA cost. A constructive lower bound on the improvement probability via selection and mutation gives $P_{\text{imp}}(t)$. Convergence follows from the monotone bounded sequence theorem. Then, apply results from Lemma 4 and Propositions 1 and 2. Taking expectations over $N_p - n_e$ independent offspring, we obtain

$$\mathbb{E}[J_{\text{EA}}(\theta_{t-1}^*) - J_{\text{EA}}(\theta_t^*)] \geq (w_c - \Phi(h_{t-1})) P_{\text{imp}}(t). \quad (19)$$

which gives the desired result. Since $J_{\text{EA}} \geq 0$ and the right-hand side is strictly positive while $h_t > h^*$, the sequence $\{J_{\text{EA}}(\theta_t^*)\}$ is hence convergent and giving $\lim_{t \rightarrow \infty} h_t \leq h^*$. $\square$

Summing over the first T active generations (i.e., those with $h_{t-1} > h^*$):

$$\mathbb{E}[J_{\text{EA}}(\theta_0^*) - J_{\text{EA}}(\theta_T^*)] \geq \sum_{t=1}^T (w_c - \Phi(h_{t-1})) P_{\text{imp}}(t). \quad (20)$$

Overall, our results in this section tell us that the EA cost will probabilistically improve until it reaches some lower limit (related to h^*), at which point it will stagnate (i.e., converge). The presented improvement probabilities (specifically, (19)) can be used to approximate cost as the EA goes through successive generations; we show in simulations (Section VII) that these quite closely match the true EA cost improvements, particularly in later generations.

V. STABILITY ANALYSIS FOR EA CO-DESIGN

In this section, we provide analysis for the closed-loop stability associated with controllers encoded by the EA population, i.e., $K_s(\theta^i)$ for $i \in \mathcal{P}_t$. The argument proceeds as follows: first, we construct a Lyapunov function for optimal LQR controller K_d and quantify its stability margin. We then relate this to the difference between the dense controller and the sparsified controller, i.e., $\Delta K := K_d - K_s$, and analyze the effects of communication link vs. actuator/sensor sparsification.

The closed-loop system (1) with dense LQR controller K_d is $A + BK_d$. By Lemma 1 and [3, Theorem A.7], this closed-loop is (Υ, ρ) -stable. A Lyapunov function $V(x) = x^\top V^* x$ for this system can be found by solving the discrete Lyapunov equation for matrix V^* . Additionally, the (Υ, ρ) -stability of this system yields

$$\|x\|^2 \leq V(x) \leq M \|x\|^2, \quad M := \frac{\Upsilon^2}{1 - \rho^2}, \quad (21)$$

Rearranging (21) gives the unit-decrease identity

$$V(x) - V(A + BK_d x) = \|x\|^2 \quad \forall x \in \mathbb{R}^{N_x}. \quad (22)$$

Theorem 2. Define the stability margin

$$\sigma_{\text{crit}} := \frac{1 - \rho^2}{4 \Upsilon^2 L (L + 2 \Upsilon \rho)}, \quad (23)$$

where L is from Assumption 1(a) and Υ, ρ are from Lemma 1. If $\sigma := \|K_d - K_s\|_F \leq \sigma_{\text{crit}}$, then the closed-loop system $A + BK_s$ is (Ω, β) -stable, where

$$\beta := \left(1 - \frac{1 - \rho^2}{2 \Upsilon^2}\right)^{1/2} \in (0, 1), \quad \Omega := \left(\frac{\Upsilon^2}{1 - \rho^2}\right)^{1/2}. \quad (24)$$

Proof. Write $A + BK_s = A + BK_d - B\Delta K$. Substituting into $V(x) = x^\top V^* x$ and applying the Lyapunov decrease

identity (22) yields

$$\begin{aligned} V(A + BK_s x) - V(x) &= -\|x\|^2 + \underbrace{(B\Delta K x)^\top V^* (B\Delta K x)}_{T_1} \\ &\quad - \underbrace{2(A + BK_d x)^\top V^* B\Delta K x}_{T_2}. \end{aligned} \quad (25)$$

We now bound terms T_1 and T_2 . For T_1 , by (21) and Assumption 1(a) ($\|B\| \leq L$),

$$T_1 \leq M \|B\|^2 \sigma^2 \|x\|^2 \leq M L^2 \sigma^2 \|x\|^2.$$

For T_2 , since $\|A + BK_d\| \leq \Upsilon \rho$ (from (Υ, ρ) -stability at time $k = 1$) and $\|V^*\| \leq M$,

$$|T_2| \leq 2 M \Upsilon \rho L \sigma \|x\|^2.$$

Combining with (25):

$$V(A + BK_s x) - V(x) \leq -\left(1 - M L \sigma (L \sigma + 2 \Upsilon \rho)\right) \|x\|^2. \quad (26)$$

When $\sigma \leq \sigma_{\text{crit}}$, we further bound these terms. First, note that

$$M L \sigma_{\text{crit}} = \frac{\Upsilon^2}{1 - \rho^2} \cdot L \cdot \frac{1 - \rho^2}{4 \Upsilon^2 L (L + 2 \Upsilon \rho)} = \frac{1}{4(L + 2 \Upsilon \rho)}.$$

Then, the quadratic term in the bound of T_1 is bounded by $M L^2 \sigma_{\text{crit}}^2 = M L \sigma_{\text{crit}} \cdot L \sigma_{\text{crit}} = \frac{1 - \rho^2}{16 \Upsilon^2 (L + 2 \Upsilon \rho)^2} \leq \frac{1}{16}$, since $\Upsilon \geq 1$, $L \geq 1$, and $1 - \rho^2 \leq 1$. The cross-term in the bound of T_2 is bounded by $2 M \Upsilon \rho L \sigma_{\text{crit}} = \frac{\Upsilon \rho}{2(L + 2 \Upsilon \rho)} \leq \frac{1}{4}$, where the inequality uses $L + 2 \Upsilon \rho \geq 2 \Upsilon \rho$. Therefore,

$$M L \sigma (L \sigma + 2 \Upsilon \rho) \leq M L^2 \sigma_{\text{crit}}^2 + 2 M \Upsilon \rho L \sigma_{\text{crit}} = \frac{5}{16} < \frac{1}{2},$$

and (26) becomes

$$V(A + BK_s x) \leq V(x) - \left(1 - \frac{5}{16}\right) \|x\|^2 \leq V(x) - \frac{1}{2} \|x\|^2. \quad (27)$$

Dividing by $V(x)$ ($x \neq 0$) and using $\|x\|^2/V(x) \geq 1/M$:

$$\frac{V(A + BK_s x)}{V(x)} \leq 1 - \frac{1}{2M} = 1 - \frac{1 - \rho^2}{2 \Upsilon^2} = \beta^2.$$

Iterating: $V(x_k) \leq \beta^{2k} V(x_0)$. Translating back via (21):

$$\|x_k\|^2 \leq V(x_k) \leq \beta^{2k} M \|x_0\|^2,$$

so $\|x_k\| \leq \Omega \beta^k \|x_0\|$ with $\Omega = M^{1/2}$. Since $\Upsilon \geq 1$ and $\rho \in (0, 1)$, we have $\beta^2 = 1 - (1 - \rho^2)/(2 \Upsilon^2) \in (0, 1)$. $\square$

Remark 1 (Interpretation of σ_{crit}). *The stability margin σ_{crit} depends on (Υ, ρ) from Lemma 1 through (23). As $\rho \rightarrow 1$ (slow spatial decay), the numerator $1 - \rho^2 \rightarrow 0$ while the denominator grows, so $\sigma_{\text{crit}} \rightarrow 0$, i.e., the allowable controller perturbation vanishes.*

Remark 2. *This Lyapunov function $V(x)$ also enables performance analysis. Since V decreases geometrically along the closed-loop trajectory with rate β^2 , standard perturbation arguments yield a sub-optimality gap of the form*

$$J_{\text{LQR}}(K_s) - J_{\text{LQR}}(K_d) = O\left(\frac{\sigma \|\Sigma\|^2}{1 - \beta^2}\right),$$

Sparse controller $K_s(\theta)$ is constructed from K_d via pruning, as detailed in previous sections. we decompose the gain perturbation from effects related to communication link pruning and actuator/sensor selection:

$$\Delta K = \underbrace{K_d - \Pi_\ell(K_d)}_{\Delta K_{\text{comm}}} + \underbrace{\Pi_\ell(K_d) - \Pi_{\mathbf{a}, \mathbf{s}}(\Pi_\ell(K_d))}_{\Delta K_{\text{as}}}, \quad (28)$$

Theorem 3. *Consider pruned controller $K_s(\theta)$ with effective truncation distance h . Define*

$$h_{\text{stab}} := \min\left\{h \geq 0 : \Upsilon \sqrt{\sum_{r>h} N_\Delta(r) \rho^{2r}} < \sigma_{\text{crit}}\right\}, \quad (29)$$

where σ_{crit} is given by (23) and $N_\Delta(r) := |\{(i, j) : d_G(i, j) = r\}|$. Then,

(i) *If $\mathbf{a} = \mathbf{1}$, $\mathbf{s} = \mathbf{1}$ and $h \geq h_{\text{stab}}$, then $A + BK_s$ is (Ω, β) -stable with Ω, β as in (24).*

(ii) *For general values of $\mathbf{a}, \mathbf{s}$, $A + BK_s$ is (Ω, β) -stable whenever*

$$\|\Delta K_{\text{comm}}\|_F + \|\Delta K_{\text{as}}\|_F < \sigma_{\text{crit}}. \quad (30)$$

Proof. First, we bound $\|\Delta K_{\text{comm}}\|_F$. Let h be the effective truncation distance of $P_{I_\ell}(K_d)$ as in Definition 2(c). By Lemma 1, every discarded entry satisfies $|K_{ij}^*| \leq \Upsilon \rho^{d_G(i, j)}$ with $d_G(i, j) > h$, so

$$\|\Delta K_{\text{comm}}\|_F^2 \leq \Upsilon^2 \sum_{r=h+1}^{\infty} N_\Delta(r) \rho^{2r}, \quad (31)$$

where $N_\Delta(r) := |\{(i, j) : d_G(i, j) = r\}|$. For bounded-degree graphs, this sum is dominated by $\rho^{2(h+1)}$, giving $\|\Delta K_{\text{comm}}\|_F \sim O(\Upsilon \rho^h)$. By (28) and the triangle inequality, $\|K_d - K_s\|_F \leq \|\Delta K_{\text{comm}}\|_F + \|\Delta K_{\text{as}}\|_F$. When $\mathbf{a} = \mathbf{1}$, $\mathbf{s} = \mathbf{1}$, $\Delta K_{\text{as}} = 0$ and Lemma 1 with (31) gives $\|\Delta K_{\text{comm}}\|_F < \sigma_{\text{crit}}$ for $h \geq h_{\text{stab}}$. Applying Theorem 2 yields (i). Part (ii) follows identically from Theorem 2. $\square$

Corollary 1 (Offspring stability probability). *At generation t , suppose the best individual has link count ℓ_t^* with $h(\ell_t^*) \geq h_{\text{stab}}$. Then any single offspring is (Ω, β) -stable with probability at least*

$$p_{\text{stab}}(t) \geq \frac{[\ell_t^* - \ell_{\text{stab}} + d + 1]_+}{2d + 1} (1 - p_m)^{N_u + N_x}, \quad (32)$$

where $\ell_{\text{stab}} := \min\{\ell : h(\ell) \geq h_{\text{stab}}\}$, $[\cdot]_+ := \max(\cdot, 0)$, and d is the mutation range. In particular, when $\ell_t^* \geq \ell_{\text{stab}} + d$, the first factor equals 1 and stability is limited only by the mask-preservation probability $(1 - p_m)^{N_u + N_x}$.

Proof. Offspring link count is $\ell_t^* + \delta$ with $\delta \sim \text{Unif}\{-d, \dots, d\}$. Stability requires $\ell \geq \ell_{\text{stab}}$ (Theorem 3(i)) and unflipped masks (probability $(1 - p_m)^{N_u + N_x}$). The two events are independent; multiplying gives (32). $\square$

In general, when the open-loop plant A is stable, this bound is satisfied by the majority of individuals in any given EA generation. However, when the open-loop plant A is unstable, this bound is violated and a substantial amount of individuals in each generation become unstable; this motivates the following section.

VI. EA MODIFICATIONS FOR UNSTABLE OPEN-LOOP PLANTS

In general, the presence of some unstable individuals is not detrimental to the EA. However, when open-loop plant A is unstable, a large portion of individuals become unstable; this renders the EA more ineffective at finding optimal solutions to (5), since most of its population genes encode solutions with infinite cost. Here, we introduce a modification to our EA algorithm to overcome this. The general idea is as follows: for each “unstable” individual i (i.e., $A + BK_s(\theta^i)$ is unstable) in the population, we develop an alternative cost evaluation mechanism. Instead of directly using controller $K_s(\theta^i)$, we introduce a repaired controller $K_s^r(\theta^i)$, which has the same sparsity as $K_s(\theta^i)$ but with different numerical values, such that $A + BK_s^r(\theta^i)$ is stable. In this way, we are able to efficiently utilize controllers encoded by genes θ^i to continue minimizing EA cost (5). The final EA output will then also be modified using this repaired controller.

Controller repair will leverage the Gershgorin disk theorem. For a matrix $M \in \mathbb{R}^{n \times n}$, define the *Gershgorin row-sum*

$$R_i(M) := \sum_{j=1}^n |M_{ij}|, \quad i = 1, \dots, n, \quad (33)$$

and the *Gershgorin radius* $\bar{R}(M) := \max_i R_i(M)$.

Lemma 5 (Gershgorin sufficient condition). *If $\bar{R}(A + BK) < 1$, then $A + BK$ is Schur stable.*

Proof. By the Gershgorin disk theorem, every eigenvalue λ of $A_{cl} = A + BK$ lies in at least one disk $\mathcal{D}_i = \{z \in \mathbb{C} : |z - A_{cl,ii}| \leq r_i\}$ where $r_i = \sum_{j \neq i} |A_{cl,ij}|$. Hence $|\lambda| \leq |A_{cl,ii}| + r_i = R_i(A_{cl}) \leq \bar{R}(A + BK) < 1$. $\square$

We are interested in improving stability without disturbing the sparsity of $K_s(\theta^i)$. To do so, we introduce set $\mathcal{K}_S(\theta) := \{K \in \mathbb{R}^{N_u \times N_x} : \text{supp}(K) \subseteq \text{supp}(K_s(\theta))\}$, i.e., the set of all controllers that have the same sparsity as $K_s(\theta)$. Roughly speaking, we will modify $K_s(\theta)$ by taking gradients to improve its stability (via Gershgorin condition) and projecting these gradients to preserve its sparsity. Let $\text{proj}_{\mathcal{K}}(K)$ denote the projection of K into $\mathcal{K}_S(\theta)$, i.e., K with all off-support entries zeroed out.

Proposition 3 (Convexity). *For each i , the function $K \mapsto R_i(A + BK)$ is convex. Consequently, the Gershgorin-stable set $\mathcal{G} := \{K : \bar{R}(A + BK) < 1\}$ is a convex (open) subset of $\mathbb{R}^{N_u \times N_x}$, and $\mathcal{G} \cap \mathcal{K}_S$ is convex.*

Proof. Fix row i . For each column j , $A_{cl,ij} = A_{ij} + \sum_{u=1}^{n_u} B_{iu} K_{uj}$ is affine in K . Therefore $|A_{cl,ij}|$ is convex in K (absolute value of an affine function). $R_i = \sum_j |A_{cl,ij}|$ is a non-negative sum of convex functions, hence convex. $\bar{R} = \max_i R_i$ is the pointwise maximum of convex functions, hence convex. The sublevel set $\{K : \bar{R}(A + BK) < 1\}$ is therefore convex. Intersecting with the affine subspace $\mathcal{K}_S$ preserves convexity. $\square$

Proposition 4. *Let $i^* = \arg \max_i R_i(A + BK)$. A subgradient of $\bar{R}(A + BK)$ with respect to K_{uj} is*

$$g_{uj} = \text{sign}(A_{i^*j} + (BK)_{i^*j}) B_{i^*u}. \quad (34)$$

Proof. Since $\bar{R}(A + BK) = \max_i R_i(K)$, a subgradient of $\bar{R}$ at K can be taken as any subgradient of R_{i^*} at K [3, Sec. 3.1.2]. Now,

$$R_{i^*}(K) = \sum_{j=1}^{n_x} |A_{i^*j} + (BK)_{i^*j}| = \sum_{j=1}^{n_x} |A_{i^*j} + \sum_u B_{i^*u} K_{uj}|.$$

Each summand $\phi_j(K) := |A_{i^*j} + \sum_u B_{i^*u} K_{uj}|$ is the absolute value of an affine function $\mu_j(K)$. When $\mu_j(K) \neq 0$, ϕ_j is differentiable with $\partial \phi_j / \partial K_{uj} = \text{sign}(\mu_j) \cdot B_{i^*u}$. When $\mu_j(K) = 0$, any value in $[-|B_{i^*u}|, |B_{i^*u}|]$ is a valid subgradient element. Summing over j gives the subgradient of R_{i^*} ; since each K_{uj} appears only in the j -th summand, the subgradient with respect to K_{uj} reduces to (34). $\square$

Scalar subgradients g_{uj} can be stacked together to form matrix subgradient g . Let $\tilde{g} := \text{proj}_{\mathcal{K}}(g)$ denote its projection to preserve sparsity. We are now ready to propose the alternative cost evaluation. Given gene θ^i where $A + BK_s(\theta^i)$ is unstable, we solve convex feasibility problem

$$\text{find } K^r \in \mathcal{K}_S \quad \text{s.t.} \quad \bar{R}(A + BK^r) \leq \rho^*, \quad (35)$$

where $\rho^* < 1$ is some target row-sum (e.g. 0.95), via the iteration

$$K^{r,(t+1)} = \text{proj}_{\mathcal{K}}[K^{r,(t)} - \eta_t \tilde{g}^{(t)}], \quad (36)$$

with Polyak step size

$$\eta_t = \frac{\bar{R}(K^{r,(t)}) - \rho^*}{\|\tilde{g}^{(t)}\|_F^2}, \quad (37)$$

Proposition 5. *If $\mathcal{G} \cap \mathcal{K}_S \neq \emptyset$ (i.e., a Gershgorin-stable controller exists on the given sparsity pattern), then the Polyak subgradient iteration (36)–(37) generates a sequence $\{K^{r,(t)}\}$ satisfying*

$$\min_{0 \leq \tau \leq t} \bar{R}(K^{r,(\tau)}) - \rho^* \leq \frac{\|K^{r,(0)} - K^*\|_F^2}{2 \sum_{\tau=0}^t \eta_\tau} \xrightarrow{t \rightarrow \infty} 0, \quad (38)$$

where $K^* \in \mathcal{G} \cap \mathcal{K}_S$ is some feasible point. In particular, the iterates reach $\bar{R} < 1$ in finitely many steps, which by Lemma 5 guarantees Schur stability.

Proof. This is a standard result for Polyak-step subgradient methods applied to convex feasibility [10]. Let K^* be any point with $\bar{R}(K^*) \leq \rho^*$. By convexity of $\bar{R}$:

$$\bar{R}(K^{r,(t)}) - \rho^* \leq \bar{R}(K^{r,(t)}) - \bar{R}(K^*) \leq \langle \tilde{g}^{(t)}, K^{r,(t)} - K^* \rangle_F.$$

From the update rule and the non-expansiveness of $\Pi_{\mathcal{S}}$:

$$\begin{aligned} \|K^{r,(t+1)} - K^*\|_F^2 &\leq \|K^{r,(t)} - \eta_t \tilde{g}^{(t)} - K^*\|_F^2 \\ &= \|K^{r,(t)} - K^*\|_F^2 - 2\eta_t \langle \tilde{g}^{(t)}, K^{r,(t)} - K^* \rangle_F \\ &\quad + \eta_t^2 \|\tilde{g}^{(t)}\|_F^2. \end{aligned}$$

Substituting the Polyak step size $\eta_t = (\bar{R}(K^{r,(t)}) - \rho^*) / \|\tilde{g}^{(t)}\|_F^2$ and using the subgradient inequality:

$$\|K^{r,(t+1)} - K^*\|_F^2 \leq \|K^{r,(t)} - K^*\|_F^2 - \frac{(\bar{R}(K^{r,(t)}) - \rho^*)^2}{\|\tilde{g}^{(t)}\|_F^2}.$$

Therefore $\{\|K^{r,(t)} - K^*\|_F^2\}$ is nonincreasing and

$$\sum_{t=0}^{\infty} \frac{(\bar{R}(K^{r,(t)}) - \rho^*)^2}{\|\tilde{g}^{(t)}\|_F^2} \leq \|K^{r,(0)} - K^*\|_F^2 < \infty.$$

Since $\|\tilde{g}^{(t)}\|_F$ is bounded (by $\|B\|_F \sqrt{n_x}$), the numerator $(\bar{R}(K^{r,(t)}) - \rho^*)^2 \rightarrow 0$, establishing fast convergence. The rate (38) follows from the standard telescoping argument for subgradient methods. $\square$

Thus, if a Gershgorin-stable controller exists on the given sparsity pattern, our method is guaranteed to find it. In practice (see Section VII), this allows us to better utilize a substantial portion of previously “unstable” genes. The method is summarized in Algorithm 2. When running general EA on an open-loop unstable system, instead of using $K_s(\theta^i)$ to evaluate cost for unstable individual i (Line 4 in Algorithm 1), use $K_s^r(\theta^i)$ instead. Similarly, when using the optimal gene to design controller (Line 12 in Algorithm 1), use controller $K_s^r(\theta^*)$ instead of $K_s(\theta^*)$. This change does not affect the asymptotic complexity or convergence properties of the original algorithm.

Algorithm 2 Alternative controller for unstable genes

Input:

Plant matrices A, B

Gene θ with $A + BK_s(\theta)$ unstable

Parameters: target row-sum $\rho^* < 1$, max iterations T

```

1:  $\mathcal{S} \leftarrow \{(u, j) : K_s(\theta) \neq 0\}$ 
2:  $K^r \leftarrow K_s(\theta)$ 
3: for  $t = 1$  to  $T$ :
4:    $A_{cl} \leftarrow A + BK^r$ 
5:   for  $i = 1$  to  $n$ :
6:      $R_i \leftarrow \sum_{j=1}^n |[A_{cl}]_{ij}|$ 
7:    $i^* \leftarrow \arg \max_i R_i$ 
8:   if  $R_{i^*} < \rho^*$ : break
9:   for  $(u, j) \in \mathcal{S}$ :
10:     $g_{uj} \leftarrow \text{sign}([A_{cl}]_{i^*j}) \cdot B_{i^*u}$ 
11:    $\eta \leftarrow \min\left(\frac{R_{i^*} - \rho^*}{\|g\|_F^2}, 0.5\right)$ 
12:    $K^r \leftarrow K^r - \eta g$ 
13: return  $K^r$ 

```

VII. SIMULATIONS

We first demonstrate the efficacy of Algorithm 1. All experiments use $Q = I_{N_x}$, $R = I_{N_u}$, cost weights $w_c = 0.05$, $w_a = 0.4$, $w_s = 0.2$, and EA parameters $N_p = 20$, $G_{\max} = 150$, $p_c = 0.8$, $p_m = 0.05$, $n_e = 10$, $\tau = 0$, $d = 5$. We test on three different plants, whose parameters are summarized in Table I. Code to reproduce simulations may be found at github.com/pengyanw/EA. The first two plants are linearized swing equations embedded in

randomized grid topologies (similar to [2]); the third plant is the same set of equations embedded in the IEEE 13-bus topology. All simulations presented in this section run on a standard laptop computer (at least, on the first author’s laptop) in about 60 seconds.

TABLE I
SIMULATION PARAMETERS FOR THE THREE EXPERIMENTS.

Parameter	5 × 5 Grid	7 × 7 Grid	IEEE 13-bus
N_x	50	98	26
N_u	25	49	13
Spec. radius of A	< 1	< 1	= 1

Results are shown in Figure 1. In our results, we include comparisons to two baselines: original (dense) LQR controller K_d and diagonal LQR (i.e., dense LQR with all cross-subsystem communication links removed)¹. We observe that for all plants, our EA always improves substantially over both baselines, reducing cost by 47–72% over dense LQR and 28–52% over diagonal LQR. Generally, dense LQR incurs high co-design penalties as it uses nearly all possible communication links, actuators, and sensors; conversely, diagonal LQR incurs high performance penalty (i.e., J_{LQR}) due to the loss of cross-subsystem communication. The EA effectively balances between these extremes. Additionally, the EA performs better compared to baseline for larger systems. We also include the numerical per-generation convergence values predicted by (19), and see that they approximate true EA behavior quite well, particularly in later generations. The optimal controller and associated communication link, actuator, and sensor selections (as returned by EA) is also shown in Figure 1; we see that actuator selection is quite sparse for all three plants. Furthermore, the controller for the IEEE 13-bus (right panel) is highly sparse, consisting of one sensor, one actuator, and one communication link.

Next, we demonstrate the effectiveness of our proposed repair mechanism on an unstable system. We use the same 5×5 grid as previously, but scale plant matrix A so that it has a spectral radius of 1.1 and is unstable. We compare the performance of Algorithm 1 alone with the performance of Algorithm 1 in combination with repair mechanism Algorithm 2, with parameter $\rho^* = 0.95$. We note that even without the repair mechanism, EA outperforms the baseline by about 25%; however, with the repair mechanism, this improvement increases to 35%. We also study the number of unstable individuals (as naively evaluated in Algorithm 1 or evaluated after repair in Algorithm 2). When no repairs occur, this value stays relatively fixed over generations; approximately half of the population is unstable at any given time. However, when repairs occur, early generations have nearly no unstable individuals. The number of unstable individuals rises in later generations, as the EA begins searching more and more sparse solutions that are more likely to be unstable prior to repair.

¹We also tested intermediate truncations of the type suggested in [3] but found that surprisingly, they are often outperformed by one of these two baselines. These are omitted for simplicity’s sake.

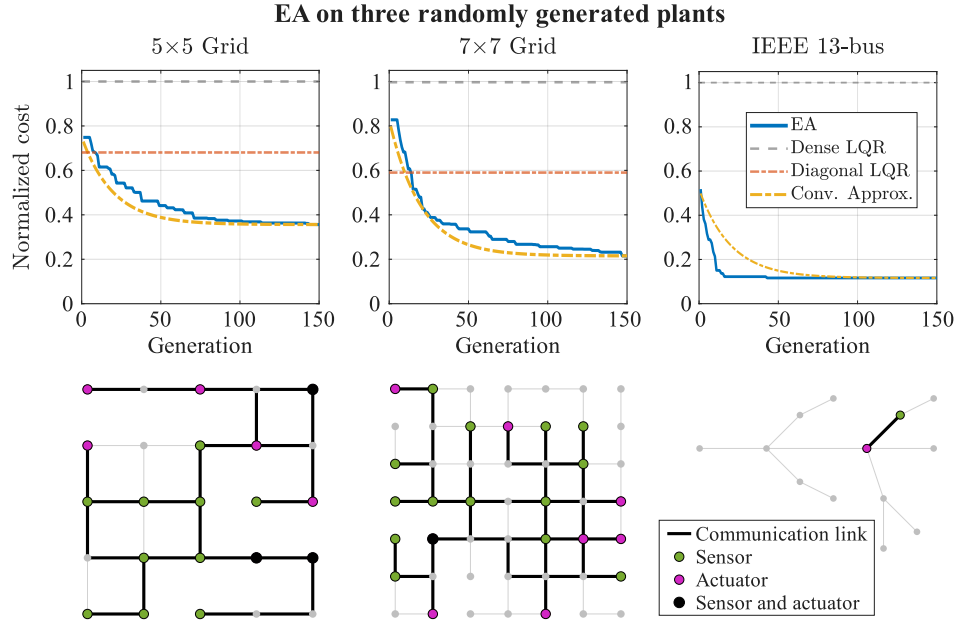


Fig. 1. Results of running Algorithm 1 on three different plants. **Top panel:** normalized cost over generations; normalized cost is defined as $J_{EA}(K^*)/J_{EA}(K_d)$, where K^* is the best per-generation controller. The solid blue line indicates EA performance; the dashed grey and red lines indicate the dense LQR and diagonal LQR baselines, respectively. For the IEEE 13-bus system, the diagonal LQR is unstable so it omitted. We also include numerical convergence approximations in the dashed yellow line using results from Section IV. **Bottom panel:** Graphical depiction of one of the optimal controllers returned by the EA at termination and its link, actuator, and sensor selections. Grey circles and dashes indicate nodes and edges in the plant. Black edges indicate communication links used by the EA controller; green and black circles indicate sensors used by the EA controller; pink and black circles indicate actuators used by the EA controller.

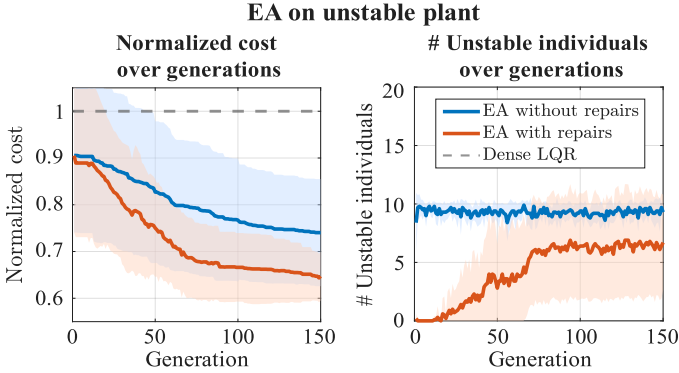


Fig. 2. Results of running Algorithm 1 (“EA without repairs”) and Algorithm 2 (“EA with repairs”) on an unstable plant, averaged over 10 different random seeds. Shaded regions indicate standard deviations. While both methods outperform dense LQR, the addition of Algorithm 2 further boosts performance.

VIII. CONCLUSIONS AND FUTURE WORK

In this paper, we proposed an evolutionary algorithm to perform co-design of LQ control cost and material cost (actuators, sensors, communication links) on a linear time-invariant plant, and demonstrated its efficacy in simulations. While this paper focuses on the LQ case, the general proposed EA framework and repair mechanism may be applicable to nonlinear systems as well; this will be the topic of future investigations.

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